SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I AB. SAI SARANYA (192126002O) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sai Saranya.ab

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	2	2	2	2	2
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	—	—

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

ANSWERS: DEBUGGING & OPTIMISATION TASK

Course: Computer Networks (CSA07) | CO2 | Assessment Tool 3

Name: **AB Sai Saranya** | Register Number: **1921260020**

Note: this task is about network configuration, so the "code" below is device configuration (Windows / Cisco IOS commands) that can be tested in Cisco Packet Tracer and checked with Wireshark. Where an output is shown, it is the typical output for that configuration.

Question 1: Workstation cannot reach the 192.168.10.0/26 subnet

Workstation settings: IP 192.168.10.65, mask 255.255.255.192 (/26), gateway 192.168.10.1. It cannot communicate with systems in 192.168.10.0/26.

1. Problem identified

With a /26 mask the block size is 256 – 192 = 64, so the subnets in this range are .0, .64, .128 and .192. I checked the address with a bitwise AND: 65 = 01000001, mask 192 = 11000000, AND = 01000000 = 64. So 192.168.10.65 belongs to **192.168.10.64/26**, not to 192.168.10.0/26 where the other systems are.

Item	Subnet 192.168.10.0/26 (other systems)	Subnet of the workstation 192.168.10.64/26
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62

Validity check. 192.168.10.65 is a valid host address by itself (it is the first usable address of 192.168.10.64/26, not the network or broadcast address). The gateway 192.168.10.1 is also a valid address, but only inside 192.168.10.0/26.

Configuration errors found

- **Error 1: wrong subnet for the IP.** The workstation is in 192.168.10.64/26, while its intended partners are in 192.168.10.0/26. Different subnets cannot talk directly at Layer 2, so the pings to those systems never work.
- **Error 2: gateway outside the host's subnet.** 192.168.10.1 is not in 192.168.10.64 – .127. A host can only use a gateway it can reach directly (it has to ARP for it), so the workstation cannot even use the router to get out.

2. Corrected configuration

Since the workstation is supposed to work with the 192.168.10.0/26 systems, I move it into that subnet and keep the existing gateway, which is the router interface:

```
C:\> netsh interface ip set address "Ethernet" static ^
        192.168.10.10 255.255.255.192 192.168.10.1

! Router side (Cisco IOS) - gateway interface of the 192.168.10.0/26 LAN
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# no shutdown
```

In Packet Tracer the same values are typed in the PC's IP Configuration window: IP 192.168.10.10, mask 255.255.255.192, gateway 192.168.10.1.

3. Explanation

192.168.10.10 gives (10 AND 192) = 0, so the network is 192.168.10.0 and the host is in the right subnet. The gateway .1 is now inside the same subnet, so the workstation can ARP for it and reach other networks. I picked .10 because .1 is reserved for the router, and it must not clash with any address already in use. Any free address from .2 to .62 would also be correct.

If the workstation really belonged to 192.168.10.64/26, the IP could stay in that subnet, but the gateway would have to be a router address in that subnet. Since .65 is the workstation's own address, the PC would move to .66 and the router interface would take .65.

4. Expected output

```
C:\> ipconfig
   IPv4 Address. . . . . . . . . . . : 192.168.10.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.192
   Default Gateway . . . . . . . . . : 192.168.10.1

C:\> ping 192.168.10.20          (another host in 192.168.10.0/26)
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
C:\> ping 192.168.10.1           (gateway)
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

5. Optimisation and result

After the correction the subnet 192.168.10.0/26 has $2^6 - 2 =$ **62 usable host addresses** (192.168.10.1 – .62). One is used by the gateway, so 61 remain for hosts. For better management, the gateway should always be the first usable address (.1) and hosts should be given addresses from a documented range or by DHCP with .1 excluded, so this kind of mistake does not repeat.

Question 2: IPv6 devices on the same network cannot communicate

Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are on the same network but communication fails.

1. Expanding both addresses

"::" stands for as many zero groups as are needed to make eight groups in total.

- **Device A:** 2001, db8, ff00, 42, 8329 are 5 groups, so "::" = 3 zero groups. Full form: **2001:0db8:0000:0000:0000:ff00:0042:8329**
- **Device B:** 2001, db8, 0, 0, 1 and 1 are 6 groups, so "::" = 2 zero groups. Full form: **2001:0db8:0000:0000:0001:0000:0000:0001**

2. Network prefix (/64 = first four groups)

Device	Groups 1 – 4 (network prefix)	Groups 5 – 8 (interface ID)
A	2001:0db8:0000:0000	0000:ff00:0042:8329
B	2001:0db8:0000:0000	0001:0000:0000:0001

Both prefixes are **2001:db8::/64** (full form 2001:0db8:0000:0000:0000:0000:0000:0000/64).

3. Same subnet? Where is the mismatch?

Yes, **both devices are in the same subnet**. Their first 64 bits are identical, both use the same prefix length /64, and their interface IDs are different, so there is no duplicate address either. So the cause is **not** an addressing or prefix mismatch, and I do not want to invent one. The only way to get a mismatch here is a misreading: if Device B is read as 2001:db8:**0:1::1** (subnet ID 1), its prefix would be 2001:db8:0:1::/64, which is a different subnet. But the address as written has the 1 in the fifth group, which is part of the interface ID.

Because the prefix is fine, the failure must come from something else that I would check in this order:

- The prefix length actually configured on the devices (for example one of them set to /128 or /48 instead of /64, or a different length on the switch or router interface).
- Layer 1/2 problems: interface down, different VLAN or wrong switch port, which stops Neighbour Discovery.
- IPv6 not enabled (or *ipv6 unicast-routing* missing on the router) and a firewall or ACL blocking ICMPv6, which IPv6 needs for address resolution.

4. Corrected / optimised configuration

The two addresses are legal, but they have odd interface IDs that are difficult to read and document. I would put both on one clean /64 with a router (gateway) at ::1:

```
! Router (default gateway of the LAN)
R1(config)# ipv6 unicast-routing
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ipv6 address 2001:db8:0:1::1/64
R1(config-if)# no shutdown

! Device A  : 2001:db8:0:1::a/64   gateway 2001:db8:0:1::1
! Device B  : 2001:db8:0:1::b/64   gateway 2001:db8:0:1::1
```

5. Expected output

```
C:\> ping 2001:db8:0:1::b
Reply from 2001:db8:0:1::b: time<1ms
```

6. Hosts available

A /64 leaves 128 – 64 = 64 host bits, so the corrected subnet supports 2^{64} = **18,446,744,073,709,551,616** addresses. IPv6 has no broadcast address, so nothing is subtracted for it. Keeping /64 is also required for SLAAC auto-configuration to work.

Question 3: Optimising the subnet plan of 192.168.1.0/24

The organization split 192.168.1.0/24 into departments with /26, /27 and /28 masks. Some departments run out of addresses while others have many unused ones. Analyse the plan, find the mistakes and recommend a VLSM scheme.

Assumptions

The task does not list how many hosts each department has, or which department got which mask. So I analyse the plan the way it is described, with one subnet of each size placed one after the other from the start of the block, and I show the method for the optimisation using example host counts that I have clearly marked as assumed. The same steps work for the real numbers.

1. Existing allocation

Subnet	Network address	Host range	Broadcast	Usable hosts	Reserved (net + bc)
/26 (255.255.255.192)	192.168.1.0	.1 – .62	192.168.1.63	62	2
/27 (255.255.255.224)	192.168.1.64	.65 – .94	192.168.1.95	30	2
/28 (255.255.255.240)	192.168.1.96	.97 – .110	192.168.1.111	14	2

2. Usable and unused addresses

- Allocated: 64 + 32 + 16 = 112 addresses; usable hosts = 62 + 30 + 14 = **106** (6 are network/broadcast addresses).
- Unallocated: 192.168.1.112 – 192.168.1.255 = 144 addresses, which is 56% of the /24 and is lying unused.

3. Inefficiencies and mistakes found

- **Sizes follow a fixed pattern, not demand.** A department needing 40 hosts on the /27 (30 usable) is short by 10, while a department with 10 hosts on the /26 wastes 52 addresses. This is exactly the complaint reported.
- **Big unmanaged leftover.** 144 addresses are free, but the departments that are short cannot use them because their blocks are already fixed and the neighbouring space is taken.
- **No room to grow.** The /28 (14 hosts) has almost no headroom, and the blocks are not planned for expansion.
- **Alignment rule to check.** Each subnet must start on a multiple of its block size (64, 32, 16). If any block is placed at a wrong start, for example a /27 at .32, it overlaps the /26 at .0 and causes duplicate addresses. In the plan above the blocks are aligned and do not overlap, so the real problem is sizing, not overlap.

4. Recommended VLSM scheme

Method: (i) list the hosts needed, largest first; (ii) choose the smallest prefix where $2^n - 2 \geq$ hosts; (iii) allocate the blocks one after another, largest first, so every block is correctly aligned. Example requirements (**assumed**): Dept A 50 hosts, Dept B 25, Dept C 12, Dept D 6, and one router-to-router link (2 hosts).

Requirement	Hosts needed	Prefix	Network	Host range	Broadcast	Usable
Dept A	50	/26	192.168.1.0	.1 – .62	.63	62
Dept B	25	/27	192.168.1.64	.65 – .94	.95	30
Dept C	12	/28	192.168.1.96	.97 – .110	.111	14
Dept D	6	/29	192.168.1.112	.113 – .118	.119	6
Router link	2	/30	192.168.1.120	.121 – .122	.123	2

Space used = 64 + 32 + 16 + 8 + 4 = 124 addresses, so **192.168.1.124 – 192.168.1.255 (132 addresses)** remains free as one continuous block for future departments. Every department fits, and the smallest ones no longer waste a /26 or /27.

```
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.192      ! Dept A
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 192.168.1.65 255.255.255.224     ! Dept B
R1(config)# interface gigabitEthernet 0/2
R1(config-if)# ip address 192.168.1.97 255.255.255.240     ! Dept C
```

5. Result

Using VLSM, each department gets a block that matches its size, utilisation rises (95 hosts needed against 114 usable here, about 83%, with the rest of the /24 kept free) and the unused space stays in one contiguous range that can be summarised and reused. With the real host counts, a department that needs growth should be given the next larger prefix, since one extra host bit doubles the block.

Question 4: IPv6 communication failure (Device A and Device B)

Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are connected to the same network but communication fails. (This is the same configuration as Question 2, so the results are the same; here I focus on how it is checked step by step.)

1. Full notation and network prefix

	Device A	Device B
Compressed	2001:db8::ff00:42:8329/64	2001:db8:0:0:1::1/64
Full notation	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000:0001:0000:0000:0001
Network prefix (/64)	2001:db8::/64	2001:db8::/64
Interface ID	::ff00:42:8329	::1:0:0:1

2. Same subnet?

Comparing the first four groups: 2001 = 2001, 0db8 = 0db8, 0000 = 0000, 0000 = 0000. All 64 prefix bits match, so the devices are on the **same subnet**, and no prefix mismatch exists in the addresses given. The only wrong reading would be taking B as 2001:db8:0:1::1, which would put it in 2001:db8:0:1::/64.

3. Debugging procedure (Packet Tracer / Wireshark)

- **Check the interfaces:** both ports up, same VLAN, and the prefix length is /64 on both devices (and on the router interface, if any).
- **Check IPv6 is enabled** and the address is not marked tentative or duplicate.
- **Capture the traffic in Wireshark** while pinging. With the filter *icmpv6.type == 135 || icmpv6.type == 136* you should see a Neighbour Solicitation from A and a Neighbour Advertisement from B. If solicitations are sent but never answered, the problem is Layer 2 or a firewall; if no solicitation is sent at all, the prefix length or routing on the sender is wrong.

```
C:\> ipconfig                      (Device A / B: check address and /64)
C:\> ping 2001:db8:0:0:1::1        (from Device A)

R1# show ipv6 interface brief      (router/switch interface state)
R1# show ipv6 neighbors            (are NS/NA entries learned?)
```

4. Corrected and optimised addressing

The addresses should be assigned in a clean and documented way in one /64:

```
Device A : 2001:db8:0:1::a/64    gateway 2001:db8:0:1::1
Device B : 2001:db8:0:1::b/64    gateway 2001:db8:0:1::1

C:\> netsh interface ipv6 add address "Ethernet" 2001:db8:0:1::a/64
```

Short sequential interface IDs are easy to read, easy to filter in Wireshark, and the /64 length keeps SLAAC working.

5. Expected output and host capacity

```
C:\> ping 2001:db8:0:1::b
Reply from 2001:db8:0:1::b: time<1ms
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

The corrected subnet has 64 host bits, so it supports 2^{64} = **18,446,744,073,709,551,616** host addresses (no broadcast address in IPv6).

Question 5: Packets for 192.168.2.100 are not reaching the intended network

Routing table: 192.168.1.0/24 via 10.0.0.2; 192.168.2.0/24 via 10.0.0.3; Default route via 10.0.0.1. Packets for 192.168.2.100 do not reach the intended network.

1. Matching entry (longest-prefix match)

A router compares the destination with every entry and picks the matching one with the longest prefix.

Entry	Range	Matches 192.168.2.100?	Prefix length
192.168.1.0/24	192.168.1.0 – .255	No	–
192.168.2.0/24	192.168.2.0 – .255	**Yes**	/24
Default (0.0.0.0/0)	everything	Yes	/0

Two entries match, but /24 is longer than /0, so the selected entry is **192.168.2.0/24** and the next-hop router is **10.0.0.3**.

2. Does the destination belong to the intended subnet?

Yes. 192.168.2.100 lies in 192.168.2.0/24 (usable range .1 – .254).

3. Routing configuration errors

As written, the table is correct: the packet is not sent to the default route or mixed up with 192.168.1.0/24, so I cannot honestly point to a wrong prefix. The fault is most likely one of these, to be checked one by one:

- **Wrong next hop for 192.168.2.0/24.** The task does not say which router really serves 192.168.2.0/24. If the networks behind 10.0.0.2 and 10.0.0.3 are swapped, the entry would have to point to the other router.
- **10.0.0.3 is unreachable:** interface down, wrong mask, or 10.0.0.3 is not in the same subnet as the router's interface. An ACL or firewall dropping the traffic is also possible.
- **Return path or onward route missing:** 10.0.0.3 may have no route to 192.168.2.0/24 or no route back, so the packet goes out but replies never return.

```
R1# show ip route 192.168.2.100
Routing entry for 192.168.2.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.0.0.3
      Route metric is 0, traffic share count is 1

R1# ping 10.0.0.3
R1# traceroute 192.168.2.100
R1# show ip arp          (ARP entry for 10.0.0.3?)
```

4. Optimised routing table

```
R1(config)# ip route 192.168.1.0 255.255.255.0 gigabitEthernet 0/0 10.0.0.2
R1(config)# ip route 192.168.2.0 255.255.255.0 gigabitEthernet 0/0 10.0.0.3
R1(config)# ip route 0.0.0.0 0.0.0.0 gigabitEthernet 0/0 10.0.0.1
```

- Giving the **exit interface together with the next hop** avoids a recursive lookup and makes the route fail faster when the link goes down.
- The two /24 networks cannot be summarised as 192.168.0.0/23 because they use **different next hops**; summarisation is only correct when routes share one next hop.
- Delete any specific route that has the same next hop as the default route (for example a /24 via 10.0.0.1); it adds nothing. If the next hops turn out to be swapped, remove the old 192.168.2.0/24 route with *no ip route* and add it again via the correct router.

5. Route used when no destination matches

If a destination matches neither 192.168.1.0/24 nor 192.168.2.0/24 (for example 8.8.8.8 or 192.168.5.10), the router uses the **default route 0.0.0.0/0 via next hop 10.0.0.1**.